

1. Title

Vendor Reliability Intelligence Platform: Vendor Reliability Intelligence & Procurement Risk Management Platform

2. Objective

Build a full-stack web application using Angular and FastAPI that enables organizations to evaluate vendor reliability, manage procurement operations, monitor supplier performance, track delivery history, maintain contract compliance, and improve procurement decision-making through centralized dashboards and analytics.

The platform helps procurement teams reduce supply chain risks by providing vendor performance insights, procurement monitoring, contract management, communication tracking, and reliability scoring.

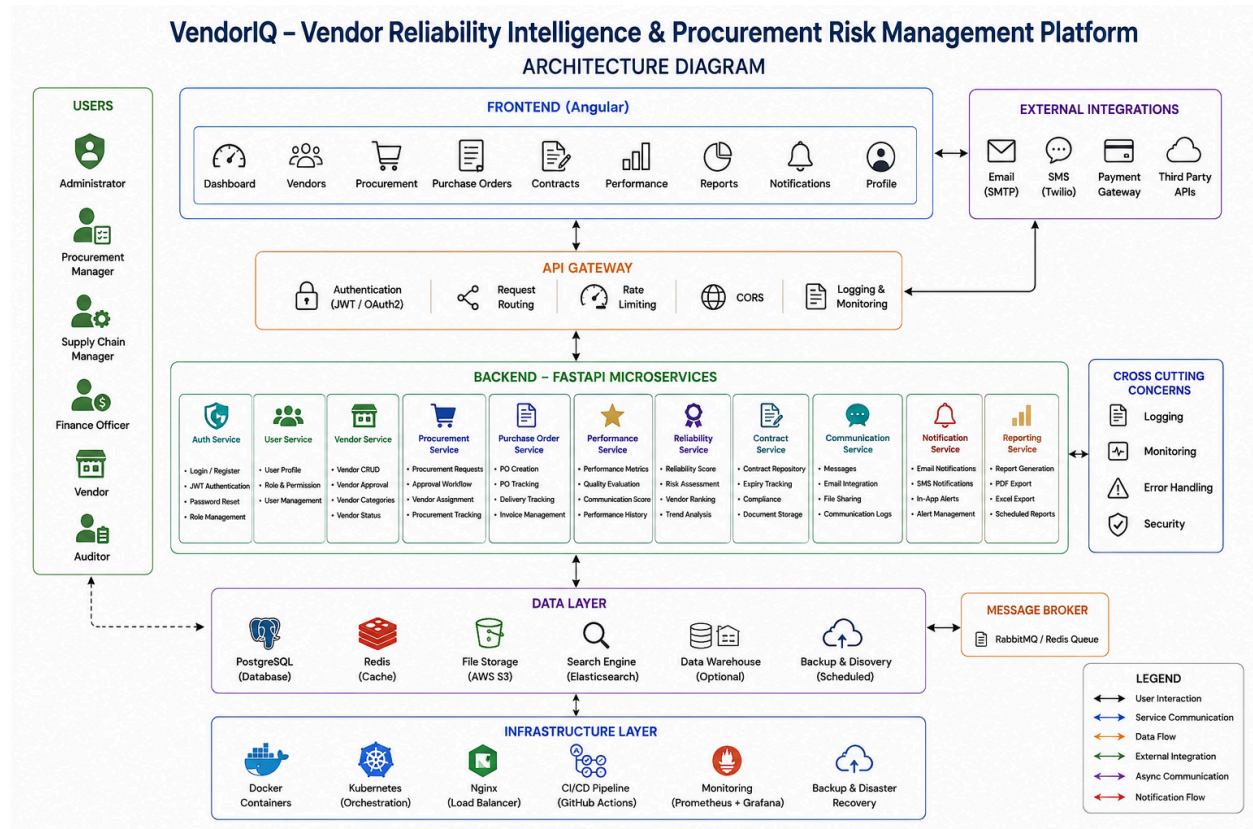
This solution can be used by manufacturing companies, retail businesses, logistics organizations, healthcare providers, construction firms, and enterprise procurement departments.

Outcomes

- (i) Developed and deployed a full-stack Vendor Reliability Intelligence platform.
- (ii) Implemented secure user authentication and role-based access control.
- (iii) Built vendor registration and supplier management modules.
- (iv) Developed procurement and purchase order management workflows.
- (v) Implemented vendor reliability scoring based on operational performance.
- (vi) Built contract and compliance monitoring systems.
- (vii) Developed dashboards for procurement analytics and vendor performance.
- (viii) Implemented notifications and procurement alerts.
- (ix) Generated procurement and vendor performance reports.
- (x) Deployed the application using Docker and cloud platforms.

(xi) Designed responsive UI wireframes and procurement user workflows before implementation.

3. Architecture Components



4. Modules to be Implemented

1. User Authentication & Role Management

Features

- (i) User Registration
- (ii) Secure Login
- (iii) JWT Authentication
- (iv) Password Reset

(v) Profile Management

(vi) Role-Based Access Control

Roles

Administrator

Procurement Manager

Supply Chain Manager

Vendor

Finance Officer

Auditor

2. Vendor Management Module

Features

(i) Vendor Registration

(ii) Vendor Profile Management

(iii) Vendor Categorization

(iv) Vendor Approval Workflow

(v) Vendor Status Monitoring

(vi) Vendor Contact Management

Vendor Categories

Raw Material Suppliers

Equipment Vendors

IT Vendors

Service Providers

Logistics Partners

Maintenance Vendors

3. Procurement Management Module

Features

- (i) Procurement Requests
- (ii) Purchase Order Creation
- (iii) Procurement Approval Workflow
- (iv) Vendor Assignment
- (v) Order Tracking
- (vi) Invoice Management

Procurement Status

Pending

Approved

Ordered

Delivered

Completed

Cancelled

4. Vendor Performance Module

Features

- (i) Delivery Performance Monitoring
- (ii) Product Quality Evaluation
- (iii) Communication Response Tracking
- (iv) Service Rating
- (v) Performance History

(vi) Vendor Ranking

Performance Metrics

On-Time Deliveries

Delayed Deliveries

Quality Rating

Response Time

Issue Resolution Time

Order Completion Rate

5. Vendor Reliability Module

Features

(i) Vendor Reliability Score

(ii) Supplier Ranking

(iii) Procurement Risk Level

(iv) Performance Trend Analysis

(v) Procurement Recommendations

Reliability Factors

Delivery History

Product Quality

Communication Efficiency

Contract Compliance

Purchase History

Issue Resolution

6. Contract & Compliance Module

Features

- (i) Contract Repository
- (ii) Contract Renewal Tracking
- (iii) Compliance Monitoring
- (iv) Certification Management
- (v) Vendor Documentation
- (vi) Contract Expiry Notifications

7. Communication Module

Features

- (i) Vendor Messaging
- (ii) Procurement Discussions
- (iii) Communication History
- (iv) Email Notifications
- (v) File Sharing
- (vi) Activity Logs

8. Dashboard & Analytics Module

Procurement Dashboard

- (i) Procurement Overview
- (ii) Active Purchase Orders
- (iii) Vendor Performance Summary
- (iv) Procurement Cost Analysis
- (v) Delivery Status

Vendor Dashboard

- (i) Vendor Performance
- (ii) Reliability Score
- (iii) Contract Status
- (iv) Order History
- (v) Communication Activity

Admin Dashboard

- (i) User Management
- (ii) Vendor Analytics
- (iii) Procurement Reports
- (iv) Compliance Monitoring
- (v) System Statistics

9. Notification Module

Features

- (i) Procurement Alerts
- (ii) Delivery Delay Notifications
- (iii) Vendor Approval Notifications
- (iv) Contract Expiry Alerts
- (v) Compliance Notifications
- (vi) Email Notifications
- (vii) SMS Notifications

10. Reports & Export Module

Features

- (i) Vendor Performance Reports

- (ii) Procurement Reports
- (iii) Purchase Order Reports
- (iv) Compliance Reports
- (v) Contract Reports
- (vi) PDF Export
- (vii) Excel Export

11. Final Integration, Testing & Deployment

5. Milestone 1: Week 1 & 2 — Requirements, UI Design, Database Design & Backend Setup

Tasks

- (i) Define project objectives.
- (ii) Gather functional and non-functional requirements.
- (iii) Design UI wireframes.
- (iv) Create user flow diagrams.
- (v) Prepare dashboard layouts.
- (vi) Design responsive screens using Figma.
- (vii) Design database schema.
- (viii) Initialize FastAPI project.
- (ix) Configure PostgreSQL.
- (x) Implement JWT Authentication.
- (xi) Create Angular frontend.

(xii) Configure Angular Material.

UI Wireframes

Login Page

Registration Page

Dashboard

Vendor Management

Procurement Dashboard

Purchase Orders

Vendor Performance Dashboard

Analytics Dashboard

Reports Dashboard

Notification Screen

Outcomes

(i) UI wireframes completed.

(ii) User flows finalized.

(iii) Database schema completed.

(iv) FastAPI backend initialized.

(v) Angular frontend initialized.

(vi) Authentication implemented.

6. Milestone 2: Week 3 & 4 — Vendor & Procurement Management

Tasks

(i) Develop Vendor Management module.

- (ii) Build Procurement Management module.
- (iii) Implement Purchase Orders.
- (iv) Develop Vendor Approval workflow.
- (v) Create Contract Management module.
- (vi) Build Communication module.

Outcomes

- (i) Vendor Management operational.
- (ii) Procurement workflow completed.
- (iii) Purchase Order module functional.
- (iv) Contract Management completed.

7. Milestone 3: Week 5 & 6 — Vendor Performance & Analytics

Tasks

- (i) Develop Vendor Performance module.
- (ii) Build Reliability Scoring module.
- (iii) Implement Analytics Dashboard.
- (iv) Create Notification System.
- (v) Generate Reports.
- (vi) Build Procurement Analytics.

Outcomes

- (i) Vendor Performance monitoring completed.
- (ii) Reliability Scoring operational.
- (iii) Analytics Dashboard functional.

(iv) Reporting module completed.

8. Milestone 4: Week 7 & 8 — Testing, Deployment & Documentation

Tasks

- (i) Perform application testing.
- (ii) Fix identified issues.
- (iii) Deploy frontend and backend.
- (iv) Configure Docker.
- (v) Prepare documentation.
- (vi) Demonstrate complete application.

Outcomes

- (i) Production-ready application.
- (ii) Deployment completed.
- (iii) Documentation completed.
- (iv) Successful project demonstration.

9. Evaluation Criteria

Milestone 1

- (i) FastAPI project setup completed.
- (ii) Angular application initialized.
- (iii) Database schema finalized.
- (iv) Authentication implemented.

(v) UI wireframes completed.

Milestone 2

(i) Vendor Management operational.

(ii) Procurement module functional.

(iii) Purchase Order workflow completed.

(iv) Contract Management functional.

Milestone 3

(i) Vendor Performance Dashboard completed.

(ii) Reliability Scoring operational.

(iii) Reports generated successfully.

(iv) Analytics Dashboard functional.

Milestone 4

(i) Application deployed successfully.

(ii) Testing completed.

(iii) Documentation prepared.

(iv) Complete workflow demonstrated.

10. Tools & Tech Stack

Programming Language

Backend

Python

FastAPI

Frontend

TypeScript

Angular

Database

PostgreSQL

Redis

Backend Framework

FastAPI

SQLAlchemy

Pydantic

Alembic

Uvicorn

Celery

Frontend Framework

Angular

Angular Material

Bootstrap

RxJS

Chart.js

Authentication

JWT Authentication

OAuth2

Role-Based Access Control

Notifications

SMTP Email

Twilio SMS

Firebase Cloud Messaging

Cloud & DevOps

Docker

Docker Compose

GitHub Actions

AWS

Azure

Nginx

Development Tools

VS Code

Git

GitHub

Postman

Figma

11. Performance Metrics

Vendor Management

Vendor Registration Success Rate

Vendor Approval Time

Vendor Response Time

Procurement Performance

Purchase Order Processing Time

Order Completion Rate

Contract Compliance Rate

Vendor Performance

Delivery Accuracy

Communication Response Time

Issue Resolution Time

Reliability Score Consistency

System Performance

API Response Time

Database Query Performance

Dashboard Loading Time

Concurrent User Handling

12. Example Quantitative Goals

Vendor Registration Time less than 3 minutes

Purchase Order Approval within 24 hours

95% On-Time Delivery Monitoring

API Response Time below 300 ms

Dashboard Load Time below 2 seconds

Support 1000+ Concurrent Users